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Lead-Free All-Inorganic Cesium Tin Iodide Perovskite for Filamentary and Interface-Type Resistive Switching toward Environment-Friendly and Temperature-Tolerant Nonvolatile Memories

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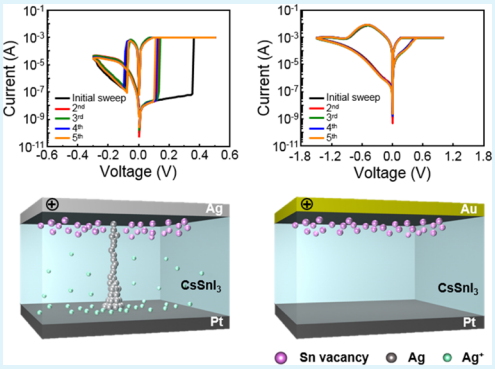
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Supporting Information

ABSTRACT: Recently, organometallic and all-inorganic halide perovskites (HPs) have become promising materials for resistive switching (RS) nonvolatile memory devices with low power consumption because they show current−voltage hysteresis caused by fast ion migration. However, the toxicity and environmental pollution potential of lead, a common constituent of HPs, has limited the commercial applications of HP-based devices. Here, RS memory devices based on lead-free all-inorganic cesium tin iodide ($CsSnI_3$) perovskites with temperature tolerance are successfully fabricated. The devices exhibit reproducible and reliable bipolar RS characteristics in both Ag and Au top electrodes (TEs) with different switching mechanisms. The Ag TE devices show filamentary RS behavior with ultralow operating voltages (<0.15 V). In contrast, the Au TE devices have interface-type RS behavior with gradual resistance changes. This suggests that the RS characteristics are attributed to either the formation of metal filaments or the ion migration of defects in HPs under applied electric fields. These distinct mechanisms may permit the opportunity to design devices for specific purposes. This work will pave the way for lead-free all-inorganic HP-based nonvolatile memory for commercial application in HP-based devices.



KEYWORDS: *lead-free halide perovskite, all-inorganic halide perovskite, resistive switching memory, electrochemical metallization, valence change mechanism*

INTRODUCTION

Resistive-switching (RS) random-access memory (ReRAM) is a promising nonvolatile memory device to complement memory-storage gaps in latency between dynamic RAM and hard disks in computing systems. It is necessary for memristors, neuromorphic computing, and logic-in-memory applications because of its low power consumption, fast switching speed, and high integration density.[1−7] Furthermore, recent progress in 3D cross-point memory devices by Intel and Micron has suggested the possibility of commercialized next-generation nonvolatile memory devices, including ReRAM.

Unlike NAND flash, a commercial nonvolatile memory device using electric charges, ReRAM stores data through current state changes under an electric field. The active layers in RS memory devices with metal/insulator/metal structures can be electrically changed from low-resistance state (LRS) to high-resistance state (HRS) by applying a voltage at different amplitudes or polarities. RS characteristics have been observed in various metal oxides such as TaO_x,[8] TiO_x,[9] and HfO_x[10] and perovskite oxides such as $SrTiO_3$,[11] $BaTiO_3$,[12] and $BiFeO_3$.[13] However, the limitations of these materials, as rigid ceramic films with poor flexibility, complicated components, and high-temperature processing requirements, have restricted further research related to facile fabrication, low-temperature processing, and flexible applications. It is essential to identify advanced materials to overcome these limitations and to meet memory performance requirements for next-generation computing system applications, such as artificial intelligence, big data, cloud computing, and the Internet of Things concept.

Organometallic and all-inorganic halide perovskites (HPs) with the formula ABX_3, where A is the organic (CH_3NH_3) or inorganic (Cs, Rb) cation, B is a metal cation (Pb, Sn), and X

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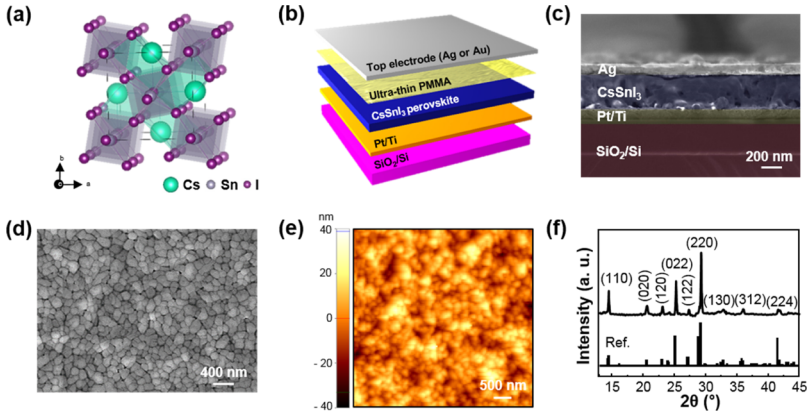


Figure 1. Memory device structure and characterization of a $CsSnI_3$ perovskite thin film. (a) Schematic of the orthorhombic structure $CsSnI_3$ perovskite phase. (b) Schematic of the Ag or Au/PMMA/$CsSnI_3$/Pt/SiO_2/Si vertical stack structure. (c) Cross-sectional SEM image of the device in false color. (d) Plane-view SEM image of the uniform and pinhole-free perovskite film. (e) AFM image of the perovskite film surface (rms = 9.352 nm). (f) XRD pattern of the perovskite film deposited on a glass substrate. Reference of the orthorhombic-structure $CsSnI_3$ from JCPDS card no. 04-014-1737.

is a halide anion (I, Br, or Cl), are considered promising functional materials because of their simple fabrication and exotic properties such as tunable band gaps, fast ion migration, facile majority carrier control, and superior flexibility, as well as their excellent optoelectronic characteristics.[14−18] Recently, progress in the HP-based RS memory devices beyond optoelectronic devices has been rapid because of their unique property of current−voltage hysteresis caused by fast ion migration.[19] RS behaviors have been observed in various HPs such as organometallic ($CH_3NH_3PbI_3$ and $CH_3NH_3PbI_{3-x}Cl_x$), all-inorganic ($CsPbI_3$ and $Cs_3Bi_2I_9$), and 2D-layered [$BA_2MA_{n-1}Pb_nI_{3n+1}$ (BA = butylammonium, MA = methylammonium)] HPs.[20−27] However, the toxicity and environmental pollution potential of lead, a common constituent of HPs, limit further research of HP-based applications. Lead-free perovskites, which contain nontoxic elements instead of lead, should be investigated more extensively for RS memory devices.

In this work, RS memory devices based on the lead-free all-inorganic HP of $CsSnI_3$ were fabricated. Unfortunately, the RS properties for this material have not yet been identified because Sn^{2+} in the HP is easily oxidized to the more stable Sn^{4+} in ambient conditions.[28,29] It has been reported that the addition of SnF_2 in the solution precursor reduces Sn^{4+} formed by oxygenated Sn^{2+}.[30] Also, the addition of SnF_2 did not distort the lattice parameters of the $CsSnI_3$ perovskite phase. Therefore, we adopted this strategy for the fabrication of the $CsSnI_3$-based RS memory device. We successfully synthesized $CsSnI_3$ by a low-temperature all-solution process and fabricated RS memory devices with structures of top electrodes (TE)/passivation layer/$CsSnI_3$/Pt/Ti/SiO_2/Si. Furthermore, temperature-tolerant all-inorganic HPs are more suitable for silicon-based commercial fabrication processes with high thermal budgets than organometallic HPs are. They exhibit bipolar RS characteristics with both Ag and Au TEs, but different switching mechanisms. The Ag TE devices show filamentary RS behavior with ultralow operating voltages (<0.15 V). In contrast, the Au TE devices have interface-type RS behavior with gradual resistance changes. To verify the distinct switching mechanisms, several analyses were conducted, including conduction mechanisms of the current−voltage (I−V) characteristics, the temperature dependence on RS, and depth profiles of elements in the $CsSnI_3$ perovskite. The temperature-tolerant all-inorganic HP-based RS memory devices were confirmed to operate over a wide temperature range compared to that of organometallic HP-based devices. It is suggested that the RS characteristics arise from either the formation of metal filaments or ion migration of defects in HPs under an electric field. We controlled the RS behaviors of the devices by applying different TEs of electrochemically active Ag and inert Au. The distinct mechanisms of the two device types may indicate the opportunity to design devices for specific purposes. This work proposes designable RS memory devices based on environment-friendly and temperature-tolerant HPs for next-generation nonvolatile memory devices.

RESULTS AND DISCUSSION

RS Characteristics. Lead-free all-inorganic $CsSnI_3$ perovskites of orthorhombic structure are successfully synthesized by a low-temperature all-solution process (Figure 1a). The vertical-stack device structure of the TE (Ag or Au)/poly(methyl methacrylate) (PMMA)/$CsSnI_3$/bottom electrode (BE, Pt)/Ti/SiO_2/Si is shown in Figure 1b. As shown in Figure 1c,d, the $CsSnI_3$ perovskite thin film of 300 nm in thickness is uniform and pinhole-free because the surface of the Pt/Ti/SiO_2/Si substrate was treated by UV ozone cleaning before the synthesis of the perovskite film. The uniform surface of the film was also confirmed by atomic force microscopy (AFM) with a root-mean-square (rms) roughness of 9.352 nm (Figure 1e). The ultrathin PMMA layer of a few nanometers in thickness on the perovskite film acts as a passivation layer, preventing the exposure of the perovskite film to oxygen and moisture under ambient conditions and reactions with the Ag TE (Figure S1, Supporting Information). The PMMA layer itself did not exhibit switching characteristics (Figure S2d) and only affected the HRS current level in the Ag TE device (Figure S2a,b). In the Au TE device, the PMMA layer was required to deposit the Au TE without degrading the perovskite film (Figure S2c). X-ray diffraction (XRD) measurements in Figure 1f show that the perovskite film is polycrystalline and single-phase with an orthorhombic structure. The obvious peaks, indexed as (110), (022), and (220) at 14°, 25°, and 29°, respectively, indicate the high crystallinity of the black orthorhombic $CsSnI_3$ perovskite phase.

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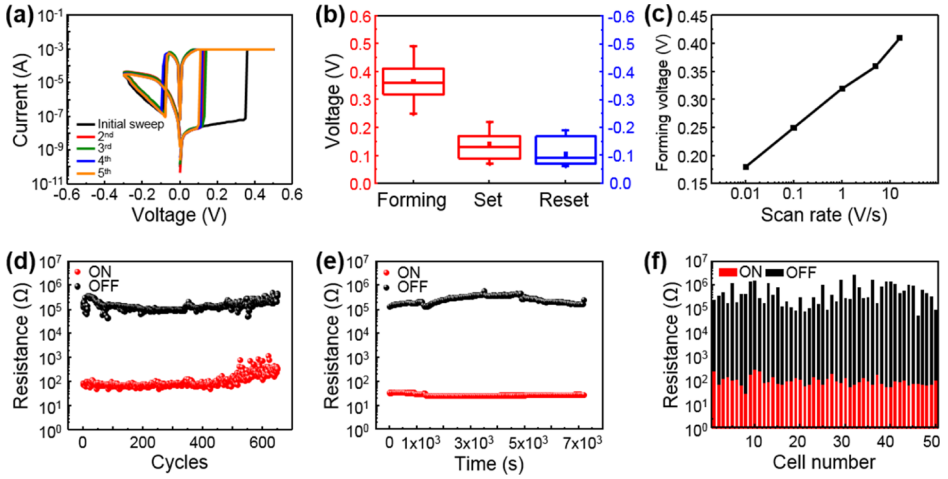


Figure 2. RS characteristics of the Ag/PMMA/$CsSnI_3$/Pt devices. (a) Series $I-V$ behaviors of the device. (b) Statistics of forming, set, and reset voltage distributions, depicted as box-whisker plots for 20 different cells. (c) Relationship between the scan rate and forming voltage in $I-V$ characteristics. (d) Reversible RS endurance under continuous write/erase voltage pulses of ±0.2 V (10 ms/20 ms pulse durations) at the read voltage of +0.05 V. (e) Retention characteristics of LRS and HRS. (f) HRS and LRS of 50 different cells. The average on/off ratio is 7×10^3.

The lead-free all-inorganic $CsSnI_3$ HP-based RS memory device with the Ag TE exhibited filamentary and bipolar RS behavior by the metallic conducting filament. To confirm the hysteresis loop of the Ag/PMMA/$CsSnI_3$/Pt device, the series $I-V$ characteristics are measured under a direct current (dc) voltage sweep of 0 V → +0.5 V → 0 V → −0.3 V → 0 V at the scan rate of 5 V s^{-1} to the TE (Ag), where the BE (Pt) is grounded (Figure 2a). During the positive-bias voltage sweep, the current compliance is set to 1 mA to prevent the irreversible breakdown of the device. The electroforming process, a kind of soft breakdown, is required in the initial sweep to the series RS, indicating a filamentary RS mechanism of the device, rather than an interface-type RS.[3] Bipolar RS abruptly occurs at +0.13 and −0.08 V after the electroforming process at +0.36 V in the series dc voltage sweeps. The initial state of the devices shows a low current (10^{-8} A), meaning that a pinhole-free perovskite thin film is successfully fabricated on the substrate by the all-solution process. During the positive-bias voltage sweep, the cell in the HRS is switched to the LRS with a high current (10^{-3} A) at the set voltage of +0.13 V (set process). The set process has a lower voltage than the electroforming process because the RS occurs at the already formed filament in the electroforming process. During the negative-bias voltage sweep, the maintained LRS is switched to the HRS at the reset voltage of −0.08 V (reset process). An ultralow electric field of 4×10^3 V cm^{-1} can also achieve the subsequent RS in the 300 nm thick perovskite thin film. Meanwhile, the current profile of HRS in the reset process is distinct to the case of the $CsPbI_3$-based RS memory device which did not exhibit RS behavior by the migration of vacancies in our previous work as shown in Figure S3.[23] The current value after reset voltage in the $CsSnI_3$-based device is much higher than that of the $CsPbI_3$-based device as shown in Figure S3a, suggesting that the Sn vacancies in the $CsSnI_3$ also have effect on the RS behavior as well as the metallic conducting filament. It is because the Sn-based HP, $CsSnI_3$, exhibits high electrical conductivity resulting from native Sn vacancies.[31] In the Ag/PMMA/$CsSnI_3$/Pt device, the Sn vacancies also affect the RS behavior, although the resistance change by the metallic conducting filament is more dominant than that by the migration of Sn vacancies due to the high conductivity of the metal. The forming and operating voltage distributions are obtained from the $I-V$ characteristics of 20 different cells to confirm the operational reliability of the device (Figure 2b). The results clearly show excellent reliability, without large variations among the different cells; the forming voltage remains larger than the set voltage. The relationship between the forming voltage and the scan rate ($\nu = dV/dt$, V = step voltage, t = step time) in the $I-V$ sweeps is confirmed in Figure 2c. The forming voltage variation according to the scan rate indicates that the resistance change is strongly dependent on the scan rate in the Ag TE device. When the scan rate is increased from 0.01 to 15.62 V s^{-1}, the point at which the resistance change occurs is increased from 0.18 to 0.41 V. Its dependence indicates that the Ag TE devices have the electrochemical metallization (ECM) mechanism by the metallic conducting filament.[2] In ECM cells, the set process by the metallic conducting filament occurs through the anodic dissolution of active metal, the migration of the metal cations across the solid electrolyte thin film, and the reduction and electrocrystallization of the metal near the BE. Under a positive electric field on Ag TE, the anodic dissolution of Ag active metal (oxidation, $Ag \rightarrow Ag^+ + e^-$) occurs. Generated Ag cations migrate to the BE by the positive electric filed on TE. When the cations reach to the grounded BE, reduction of the cations ($Ag^+ + e^- \rightarrow Ag$) occurs and the Ag conducting filament is formed by electrocrystallization of Ag metal. The dependence of the switching voltage V on the scan rate ν is described as the following equation

$$V = \frac{kT}{\alpha z e} \ln \nu + \frac{kT}{\alpha z e} \ln \frac{Q \alpha z e}{i_0 \pi r_f^2 kT} \quad (1)$$

where V is the set voltage, ν is the scan rate in $I-V$ sweeps, α is the cathodic charge-transfer coefficient, z is the oxidation number of the metal, i_0 is the electrodeposition current density, r_f is the radius of the metal filament, and Q is the required charge for the 1D growth of the metal filament up to the set switching.[32] To confirm the RS by the electrochemical reaction of the metal TE, voltage was applied at negative bias to the initial cell, and no switching characteristic was observed (Figure S4). This indicates that the Ag cations, dissolved by the positive-voltage bias on the Ag TE, form the conducting

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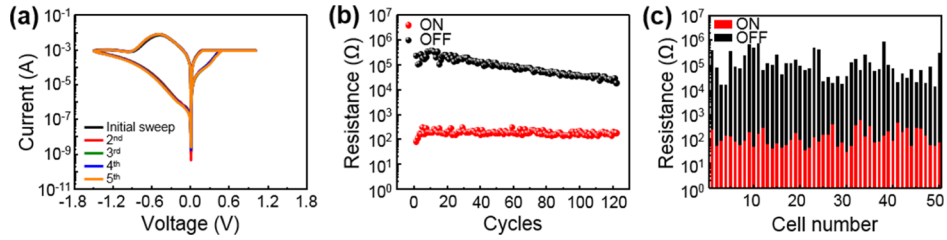


Figure 3. RS characteristics of the Au/PMMA/$CsSnI_3$/Pt devices. (a) Series $I-V$ behaviors of the device. (b) Reversible RS endurance of the device under continuous $I-V$ sweeps at the read voltage of −0.15 V. (c) HRS and LRS of 50 different cells. The average on/off ratio is 1×10^3.

filament in the perovskite layer by migration, reduction, and electrocrystallization near the BE. RS by alternating current (ac) voltage pulses is also possible for the Ag/PMMA/$CsSnI_3$/Pt device (Figure 2d). The reversible RS (endurance) is measured for 600 cycles with the on/off ratio over 10^3 under continuous write/erase voltage pulses of ±0.2 V (10 ms/20 ms pulse durations) at the read voltage of +0.05 V. The typical retention characteristics show that the HRS and LRS are maintained for up to 7×10^3 s at the read voltage of +0.02 V (Figure 2e). The HRS and LRS values of 50 different cells are confirmed under write/erase ac voltage pulses; the average on/off ratio is 7×10^3, without considerable deviation among the cells (Figure 2f). This shows the reliability and reproducibility of the devices for all-inorganic HP-based RS memory devices.

As shown in Figure 3a, the series $I-V$ characteristics of the Au/PMMA/$CsSnI_3$/Pt device show interface-type and bipolar RS behavior by the migration of Sn vacancies, unlike the Ag TE device. The series $I-V$ curves are measured under a dc voltage sweep of 0 V → −1.5 V → 0 V → +1 V → 0 V at the scan rate of 5 V s^{-1} to the TE (Au), where the BE (Pt) was grounded. They exhibit a gradual RS in contrast to the abrupt RS in the Ag TE device, indicating that the devices have different switching mechanisms depending on the TE. The contact resistance between TE and perovskites decreases when the Ag TE was replaced by Au TE because Au metal has a higher work function of 5.1 eV than Ag metal with a work function of 4.52 eV. The current value of HRS in the Au TE device showed much higher value than that of the Ag TE device. In the Au TE, an inert metal, RS behavior is caused by the Sn vacancies, indicating the valence change mechanism (VCM) in the device. Sn vacancies have the lowest formation energy among the defects in $CsSnI_3$; these create highly mobile holes.[31] The gradual current change under the applied voltage in the $I-V$ sweeps indicates interface-type switching, in which the RS occurs at the interface between the metal electrode and the perovskite thin film by the Sn vacancy migration.[3] In the p-type $CsSnI_3$ perovskite, the positive bias causes accumulation of Sn vacancies at the interface and decreases the depletion width in the perovskite layer. The resistance changes to the LRS as the contact resistance is decreased (set process). Applying a negative bias decreases the number of vacancies at the interface and leads to the HRS because of the increased depletion width in the perovskite layer and increased contact resistance (reset process). In the Au TE device, the depletion region at only upper interface in the perovskite has a dominant role for interface-type gradual RS because the Pt BE has Ohmic contact to the perovskite. The $I-V$ curves represented asymmetrical bipolar RS properties. It arises from the electrical conductivity of $CsSnI_3$ and retention failure in the Au/PMMA/$CsSnI_3$/Pt device as shown in Figure S5. $CsSnI_3$ is known as a metallic perovskite resulting from Sn vacancies which arise intrinsically.[31] There are native Sn vacancies at the interface regardless of the previous resistance state due to retention failure in the device. Therefore, the density variation of vacancies is higher under the negative bias voltage sweep than the positive bias voltage sweep because of the intrinsic density state of vacancies, which results in asymmetrical bipolar RS properties in the Au/PMMA/$CsSnI_3$/Pt device. Endurance is demonstrated over 120 cycles with the on/off ratio over 5×10^2 under a continuous dc voltage sweep of 0 V → −1.5 V → 0 V → +1 V → 0 V at the scan rate of 5 V s^{-1} and the read voltage of −0.15 V (Figure 3b). To show the reliability and reproducibility of the devices, the HRS and LRS values of 50 different cells are confirmed under dc voltage sweeping; the average on/off ratio is 1×10^3 without considerable deviation among the cells (Figure 3c). In the Au/PMMA/$CsSnI_3$/Pt device, retention failure occurs because the accumulated vacancies at the interface are quickly diffused after the voltage is removed.[33] To maintain the retention requirements for ReRAM, the retention performance can be improved by changing the BE from Pt to indium tin oxide (ITO). The memory devices with Ag or Au TE showed bipolar RS behavior, which means that a Schottky barrier must have existed between TE and the perovskite. The BE, Pt metal, which has a higher work function of 5.93 eV than Ag and Au of 4.52 and 5.1 eV, has Ohmic contact to the perovskite. Therefore, the concentration of Sn vacancies at only the upper interface in the perovskite affects the RS behavior in the Au/PMMA/$CsSnI_3$/Pt device. When the Pt BE is replaced by ITO with a work function of 4.7 eV, Ohmic contact between BE and the perovskite is converted to Schottky contact. Depletion region in the perovskite exists on both interfaces contacting the TE and BE in Au/PMMA/$CsSnI_3$/ITO. As a result, the Sn vacancies at interfaces in the perovskite no longer has effect on the RS behavior. The Au/PMMA/$CsSnI_3$/ITO device showed filamentary and bipolar RS with improved retention properties, which are different from the Au/PMMA/$CsSnI_3$/Pt device of interface-type RS behavior. The RS characteristics of the Au/PMMA/$CsSnI_3$/ITO device were confirmed as shown in Figure S6. $I-V$ characteristics were measured under the dc voltage sweep of 0 V → +2 V → 0 V → −2 V → 0 V at the scan rate of 5 V s^{-1} to the TE (Au), where the BE (ITO) was grounded. They showed filamentary and bipolar RS characteristics caused by the migration of Sn vacancies (Figure S6a). To confirm the conduction mechanisms for the HRS and LRS, $I-V$ sweeps were replotted on a double-logarithm scale (Figure S6b). They exhibited Ohmic conduction and space charge-limited conduction (SCLC) in HRS, indicating that the VCM was dominant in the device.[25] Under an electric field, a conducting filament composed of Sn vacancies is formed and results in resistance changes. The endurance and retention characteristics were confirmed over 150 cycles and 10^4 s under

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ac voltage pulses of +2 V/−3 V (10 ms pulse duration) at the read voltage of +0.05 V (Figure S6c,d).

Conduction and Switching Mechanisms. The conduction mechanisms were analyzed to describe the $I-V$ characteristics of the HRS and LRS. The typical $I-V$ curves were replotted on a double-logarithm scale to confirm the electrical transport behaviors of the devices, as shown in Figure 4. Despite the identical $CsSnI_3$ perovskite active layers in the

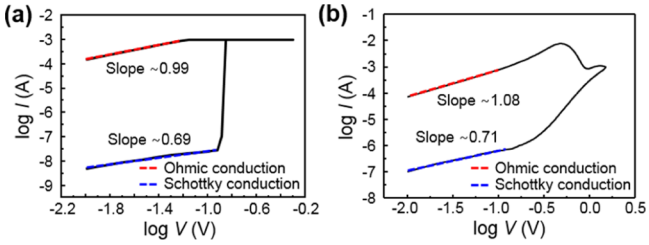


Figure 4. Conduction mechanisms of LRS and HRS. (a,b) Double-logarithm plot of the $I-V$ characteristics of the (a) Ag TE during the set process and (b) Au TE devices during the reset process.

RS devices, the conduction mechanisms differ depending on the TE. During the set process under the dc voltage sweep of 0 V → +0.5 V → 0 V on the Ag TE, the region of HRS conforms to Schottky emission $[\ln(I) \propto V^{1/2}]$ with a slope of 0.69; the LRS region shows linear Ohmic conduction ($I \propto V$) with a slope of 0.99 (Figure 4a). They were replotted as a linear fit line to verify Schottky emission and Ohmic conduction for each state (Figure S7). This indicates the ECM mechanism, where the metal cations from the dissolution of the active TE migrate to the BE and form the conducting filament. In the HRS, carrier movement is blocked by the Schottky contact between the Ag TE and the perovskite thin film. After forming the metallic conducting filament within the perovskite layer, Ohmic conduction is dominant in the LRS. In the Au TE device, $I-V$ sweep of the reset process represented the Schottky conduction in HRS more clearly than that of the set process due to asymmetrical bipolar RS properties. During the reset process under dc voltage sweep of 0 V → −1.5 V → 0 V on the Au TE, the region of HRS conforms to Schottky emission $[\ln(I) \propto V^{1/2}]$ with a slope of 0.71; the LRS region shows linear Ohmic conduction ($I \propto V$) with a slope of 1.08 (Figure 4b). The gradual RS behavior was confirmed from Schottky emission to Ohmic conduction in the Au TE device. This supports the dominance of interface-type RS by VCM in this device. The resistance state of the device gradually changes from HRS to LRS through the accumulation of Sn vacancies at the interface.

The temperature dependence of the $I-V$ characteristics of the Ag TE device is measured over the temperature range 113−373 K (Figure 5). As shown in Figure 5a, the HRS current value before the forming process increases with the temperature, which is replotted to extract the activation energy (E_a) in Figure S8. The operating voltages in the $I-V$ characteristics of the Ag TE device depend on the temperature because RS by the metallic conducting filament is caused by electrochemical reactions and thermal effects.[34,35] Figure 5b,c exhibits the tendency of operating voltages under the dc voltage sweep according to the temperature variation. The $I-V$ characteristics were plotted in linear scale to reveal the distribution of forming, set, and reset voltages more clearly in Figure S9. The voltages and temperature are inversely proportional, verifying the dominant ECM mechanism in the Ag TE device.[23] In the ECM mechanism, the forming and set processes occur by the dissolution of metal, the migration and reduction of the metal cations, and the inhomogeneous nucleation of the metal near the counter electrode. Therefore, the migration speed of the metal cations and the supersaturation of the metal near the counter electrode determine the forming and set voltages. At high temperatures, Ag cations can migrate more easily, allowing them to reach supersaturation more quickly. As the temperature increases in the heterogeneous nucleation process, it is possible to achieve supersaturation at a low voltage, thus reducing the forming and set voltages.[36] The reset process, by the rupture of the filament, is related to Joule heating-assisted dissolution. Thus, the thermal stability of the formed metal filament affects the reset process. At high temperature, filament rupture can occur more easily, and the reset process is possible under a low electric field.[37]

To reveal the VCM in the Au TE device, the $I-V$ characteristics of the HRS current value over the temperature range 273−373 K are replotted as an Arrhenius plot (Figure 6a). In the HRS, the current flows through thermally excited

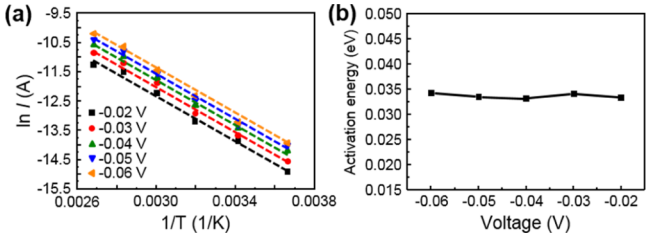


Figure 6. Temperature dependence on $I-V$ characteristics of the Au/PMMA/$CsSnI_3$/Pt device over the wide temperature region (273−373 K). (a,b) Arrhenius (ln I vs $1/T$) plot (a) and extracted activation energy (b) at each voltage for electron transport in HRS.

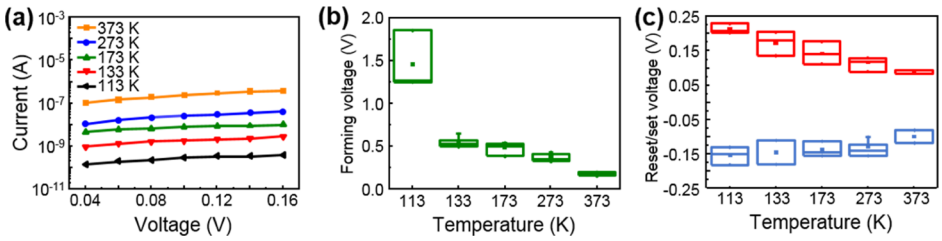


Figure 5. Temperature dependence on $I-V$ characteristics of the Ag/PMMA/$CsSnI_3$/Pt device over the temperature range (113−373 K). (a) $I-V$ characteristics of HRS in initial cell. (b,c) Statistics of forming voltage (b) and set and reset voltage (c) distributions in $I-V$ sweeps, depicted by box-whisker plots.

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electrons hopping via shallow trap states.[38] E_a for electron transport is extracted as approximately 0.033 ± 0.0006 eV, which suggests that Sn vacancies act as trap sites for electron-hopping conduction (Figure 6b).[39] This is distinct from the HRS tendency in the Ag TE device (Figure S8). The extracted E_a of the Au TE device is 6 times lower than that of the Ag TE device, resulting from Sn vacancies acting as trap sites for electron-hopping conduction. Meanwhile, both Ag and Au TE devices showed temperature tolerance over a wide temperature region from 113 to 393 K (Figure S10). It exhibits that all-inorganic HP-based devices have a potential strength in silicon-based commercial fabrication processes for high-thermal budgets compared to organometallic HPs.

Time-of-flight secondary-ion mass spectrometry (ToF-SIMS) is used to examine the depth profile of the Ag distribution within the $CsSnI_3$ perovskite layer to confirm the Ag cation migration (Figure 7). ToF-SIMS analysis is

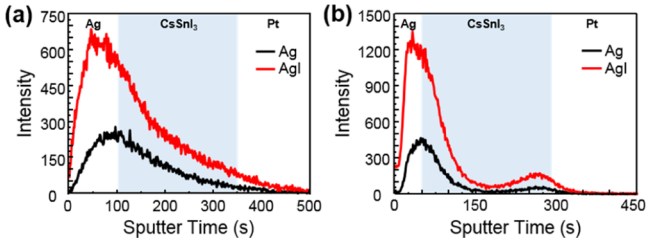


Figure 7. ToF-SIMS depth profiles from the TE of Ag/PMMA/$CsSnI_3$/Pt device. (a) Initial cell before RS. (b) Operated cell after RS.

performed using a sample with the same structure as the actual memory device. Sputtering was performed from the Ag TE in a square area of 100 μm × 100 μm; ToF-SIMS depth profiles are analyzed in the regions of interest measuring 35 μm × 35 μm aligned with the Ag TE. The ion intensity profiles of Ag and AgI are plotted together because the detected intensity of AgI, formed by highly reactive iodine, is identical to that of Ag with higher intensity. The analysis was performed on both the initial and operated cell to examine the element composing the conducting filament in RS. The boundaries of each layer were separated by the intensity of the Ag and Pt ions (Figure S11). The depth profiles of the Ag and AgI ions were distinct at the interface between the perovskite and the Pt BE when comparing the initial cell with the operated one. For the initial cell, the depth profile of the ions gradually changes without fluctuation (Figure 7a). However, that of the operated cell exhibits a fluctuating Ag distribution, indicating that Ag is diffused to the BE after applying positive bias voltage on TE and thus forms the conducting filament within the perovskite layer (Figure 7b). This supports the ECM-dominant RS mechanism in the Ag TE device, where the RS is caused by the metallic conducting filament formed by the electrochemical reaction of Ag TE. ToF-SIMS analysis of Au TE devices was also conducted; no difference in Au distribution occurred for the initial and operated cells (Figure S12). As expected, it is found that the VCM-dominant RS mechanism is not related to the electrochemical reaction of the Au TE.

A proposed filamentary ECM mechanism of the Ag/PMMA/$CsSnI_3$/Pt device is illustrated in Figure 8. When a positive bias voltage is applied to the Ag TE in the initial cell, sequential processes occur according to the anodic dissolution of the active metal, the migration of metal cations to BE, and reduction and electrocrystallization of the metal near the BE. Above the forming voltage, the HRS of the device is converted to the LRS by the metallic conducting filament formed between the TE and BE within the $CsSnI_3$ perovskite layer. By applying the electric field in the opposite direction, the filament is disrupted by Joule heating-assisted dissolution and the LRS of the device is changed to the HRS. After the forming process, the reversible RS occurs by repeating the set and reset processes under the opposite voltage bias. Interface-type VCM of the Au/PMMA/$CsSnI_3$/Pt device is suggested as described in Figure 9. The energy band diagram exhibits the Schottky

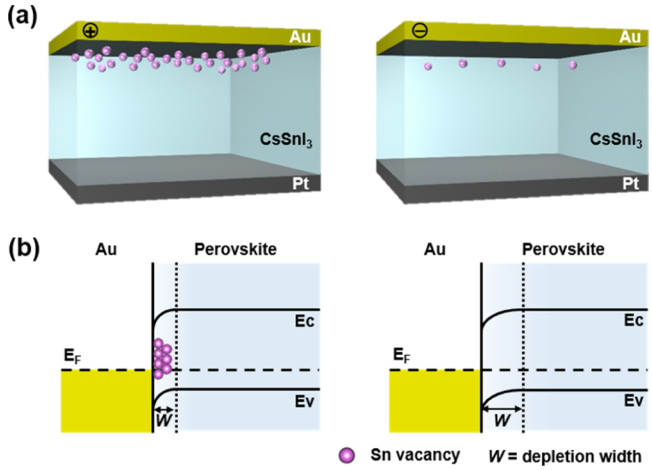


Figure 9. (a) Schematic of the mechanism in the Au/PMMA/$CsSnI_3$/Pt device. (b) Depletion width variation in the p-type perovskite layer according to the accumulation of Sn vacancies under an electric field.

barrier with the depletion region between the Au TE and p-type perovskite. Contact resistance originates in the Schottky barrier and can be changed by the applied electric field. Under a positive voltage bias on the Au TE, Sn vacancies are accumulated at the interface between the TE and perovskite layer; these reduce the depletion width in p-type perovskite layer, as shown in the band diagram. As the depletion width which corresponds to the potential profile of barrier decreases,

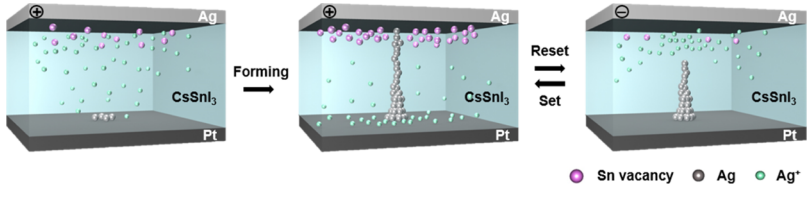


Figure 8. Schematic of the mechanism in the Ag/PMMA/$CsSnI_3$/Pt device. After the forming process, continuous set and reset processes occur by the formation and rupture of the conducting filament under an electric field.

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ACS Appl. Mater. Interfaces 2019, 11, 8155−8163

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electrons easily pass through the thin Schottky barrier by a tunneling process. Thus, the resistance state is gradually changed to the LRS as the contact resistance is decreased by the narrowed depletion width. Applying the opposite voltage bias decreases the Sn vacancies at the interface and induces the HRS by increasing the depletion width in the p-type perovskite layer and contact resistance.

In this work, the RS switching characteristics were clearly distinct depending on the TEs (Ag or Au). Although each mechanism (ECM and VCM) has remarkable properties, they have tradeoffs in strengths and weaknesses. The VCM-dominant RS is caused by the ion migration of defects in the active layer. The VCM cells generally show high switching speeds and superior retention properties but also require high electric fields because of the defects to overcome the large diffusion barrier in the active layer.[40,41] Meanwhile, the ECM-dominant RS is induced by the migration of metal cations from the dissolved electrochemically active electrode. ECM cells can be operated at ultralow voltages and low currents with high on/off ratios, but it is difficult to maintain the resistance state after removing the applied electric field.[42] It is noted that the $CsSnI_3$-based RS memory devices can be operated by both the filamentary ECM mechanism and interface-type VCM. These distinct mechanisms can provide an opportunity to design memory devices for a specific purpose by adopting the preferred characteristics of each mechanism.

CONCLUSIONS

The lead-free all-inorganic HP of $CsSnI_3$ is successfully synthesized and RS memory devices of TE (Ag or Au)/PMMA/$CsSnI_3$/Pt/SiO_2/Si are fabricated by using different TE materials of electrochemically active Ag and inert Au. They exhibit bipolar RS characteristics with both Ag and Au TEs with different switching mechanisms. We propose that the RS characteristics arise from either the formation of metal filaments or ion migration of defects in the HP under an applied electric field. The distinct switching behaviors provided by the filamentary ECM mechanism and interface-type VCM can permit the design of devices for specific use purposes. This work demonstrates designable RS memory devices based on environment-friendly and temperature-tolerant HPs for commercial applications.

EXPERIMENTAL SECTION

Materials. Tin(II) iodide (SnI_2, ultra-dry, 99.999%) and hydroiodic acid (57% in aqueous solution, stabilized with 1.5% hypophosphorous acid) were purchased from Alfa Aesar. Cesium iodide (CsI, 99.9%), tin fluoride (SnF_2, 99%), PMMA (acrylic), *N*,*N*-dimethylformamide (DMF, anhydrous, 99.8%), dimethyl sulfoxide (DMSO, anhydrous, ≥99.9%), and chlorobenzene (anhydrous, 99.8%) were sourced from Sigma-Aldrich.

$CsSnI_3$ Solution Preparation. The $CsSnI_3$ precursor solution was prepared by stirring CsI, SnI_2, and SnF_2 (1:1:0.2 mol %) powders in anhydrous DMF and DMSO (95:5 wt %) at 40 wt % for 10 min at 100 °C under a N_2 atmosphere. Hydroiodic acid (6 vol %) was added to the precursor solution to enhance the solubility of the precursors and the stability of the perovskite phase.

Fabrication of the RS Memory Device. Pt/Ti-coated SiO_2/Si substrates were prepared by electron beam evaporation. A 20 nm thick Ti layer was deposited for better adhesion of the subsequently deposited 50 nm thick Pt layer. The Pt/Ti/SiO_2/Si substrates were sequentially cleaned using acetone, isopropanol, and deionized water under ultrasonication and treated by UV−ozone for 15 min before synthesizing the $CsSnI_3$ perovskite thin film. The prepared precursor solution was spin-coated on the substrates at the spin-coating rate of 4000 rpm for 30 s and annealed at 100 °C for 3 min. After cooling to room temperature, a solution of PMMA in chlorobenzene (5 mg mL^{-1}) was sequentially spin-coated on the $CsSnI_3$ perovskite thin film at 4000 rpm for 30 s and then annealed at 100 °C for 3 min. Finally, TEs of 50 μm × 50 μm were deposited on the PMMA/$CsSnI_3$ film by e-beam evaporation at the pressure of 1 × 10^{-6} Torr and room temperature using a shadow mask.

Characterization. XRD measurement was recorded at room temperature in the 2θ range 10°−50° with a step size of 0.02° and a scan speed of 10° min^{-1} by an X-ray diffractometer (Bruker Miller Co., D8-ADVANCE) with Cu Kα radiation (λ = 1.54056 Å). The $CsSnI_3$ thin-film surface images and cross-sectional images of the device were obtained using a field-emission scanning electron microscope (ZEISS, MERLIN Compact) with an in-lens secondary electron detector at 1 kV accelerating voltage. AFM (Park Systems XE100) was used to determine the topography of the $CsSnI_3$ thin film synthesized on the Pt/Ti/SiO_2/Si substrate. Depth profiles were measured by ToF-SIMS-5 (IONTOF, Germany) with a Cs 1 keV sputter gun. The electrical properties were measured by an Agilent 4156C semiconductor analyzer in the direct current voltage-sweeping mode and alternating voltage pulse mode in a vacuum chamber (6 × 10^{-2} Torr).

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsami.8b15769.

Optical microscope images, *I*−*V* characteristics of Ag/PMMA/$CsSnI_3$/Pt devices, RS characteristics of Au/PMMA/$CsSnI_3$/ITO devices, linear-fit line of *I*−*V* characteristics, temperature dependence of *I*−*V* characteristics, temperature-tolerant characteristics, and ToF-SIMS depth profiles (PDF)

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Notes

The authors declare no competing financial interest.

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